Name of the Faculty	**Dr.M.Usha**
Department	**Information Technology**
Designation	**Dean – School of Computational Sciences**
Name of the Course	**Database Management System**
Semester	**IV**
Course Code	**24BIT43C**
Batch	**2024**
Unit & Course Topic	**Database Concepts**

UNIT I
DATABASE CONCEPTS

SYLLABUS

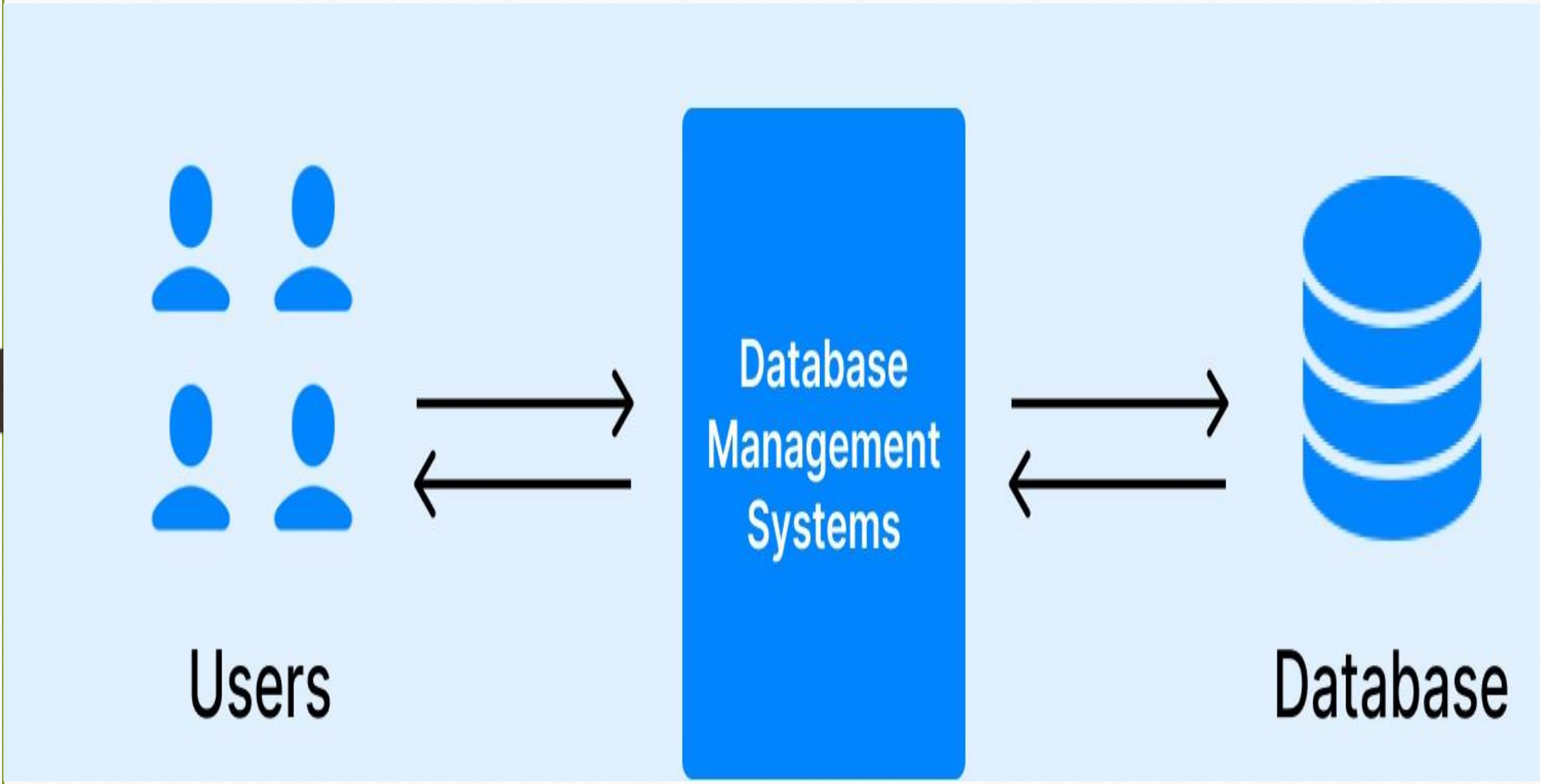
Database Concepts: Database systems - Data vs Information - Introducing the database -File system - Problems with file system.

Data models: Importance – Basic building blocks - Business rules - Evolution of data models - Degrees of data abstraction.

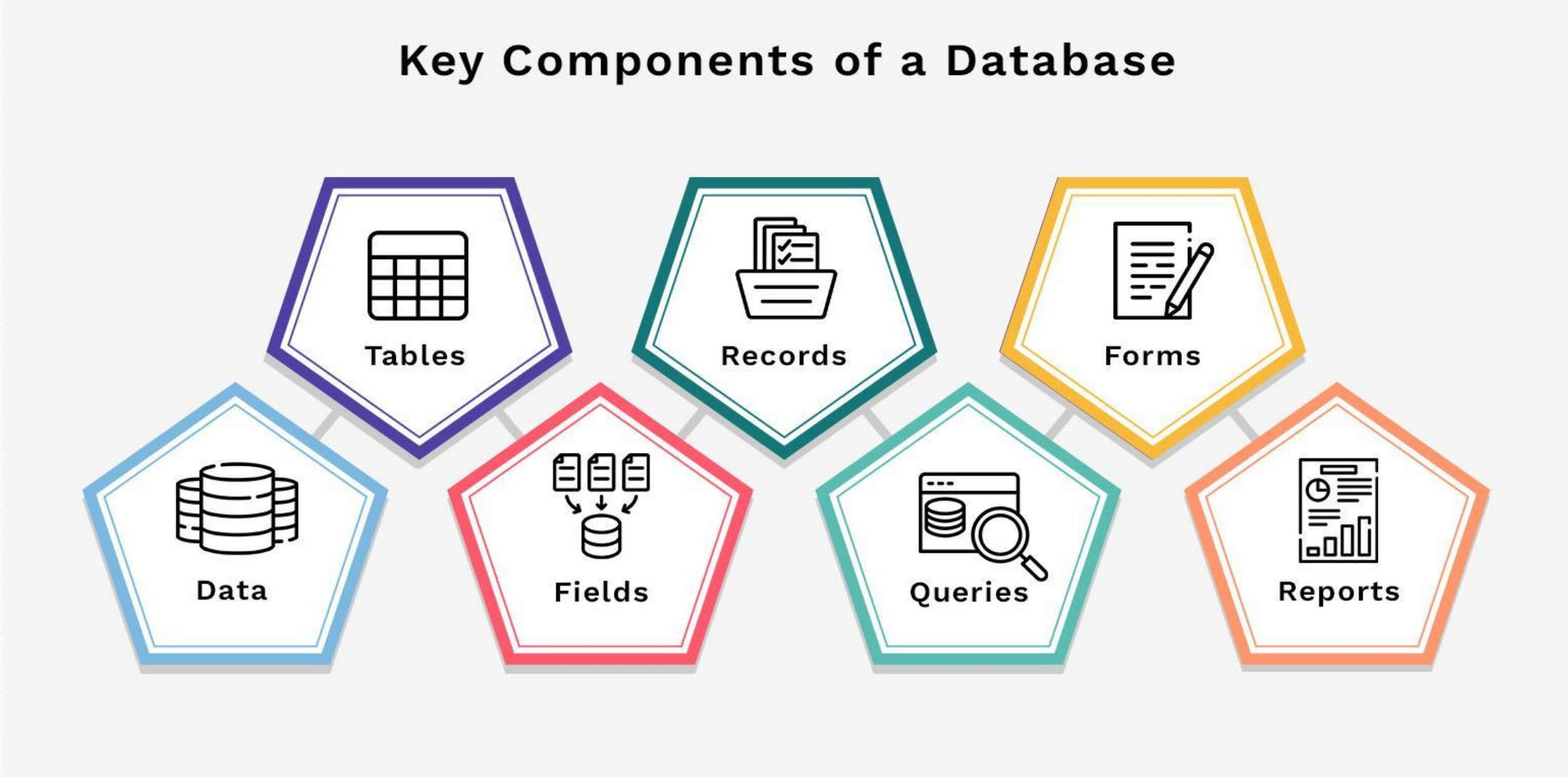
DATABASE SYSTEMS

- A database system is a collection of software that manages a structured collection of data, known as a **database**.
- The core component is the **Database Management System (DBMS)**, which acts as an intermediary between users/applications and the data, handling tasks like defining, storing, retrieving, and updating information securely.

Database Management Systems



Key Components of a Database



Key Components of a Database

Key components and functions

- **Database Management System (DBMS)**.
- **Database:** A structured collection of data, which can be in various formats (text, images, etc.).
- **Data storage:** A DBMS organizes and stores data in a structured format, making it efficient to store and retrieve
- **Data manipulation:** It provides an easy and effective way to manipulate data, such as inserting, updating, and deleting information.
- **Security and integrity**: The DBMS protects data from unauthorized access, system crashes, or data theft and ensures data consistency and integrity.
- **Concurrency control:** It allows multiple users to access the database at the same time without interfering with one another.
- **Backup and recovery**: It facilitates administrative tasks like backups and recovery to prevent data loss.

Types of Database Systems

1. **Relational Database Management Systems (RDBMS):**

These are the most common type, storing data in tables with rows and columns. They use SQL (Structured Query Language) to interact with the data. Examples include **MySQL, Oracle, and PostgreSQL.**

2. NoSQL databases:

A newer category designed to handle large-scale, unstructured, or semi-structured data. Examples include **MongoDB and Cassandra.**

Data Vs Information

Data : Data is a raw and unorganized fact that is required to be processed to make it meaningful. It can be considered as facts and statistics collected together for reference or analysis.

Characteristics of Data

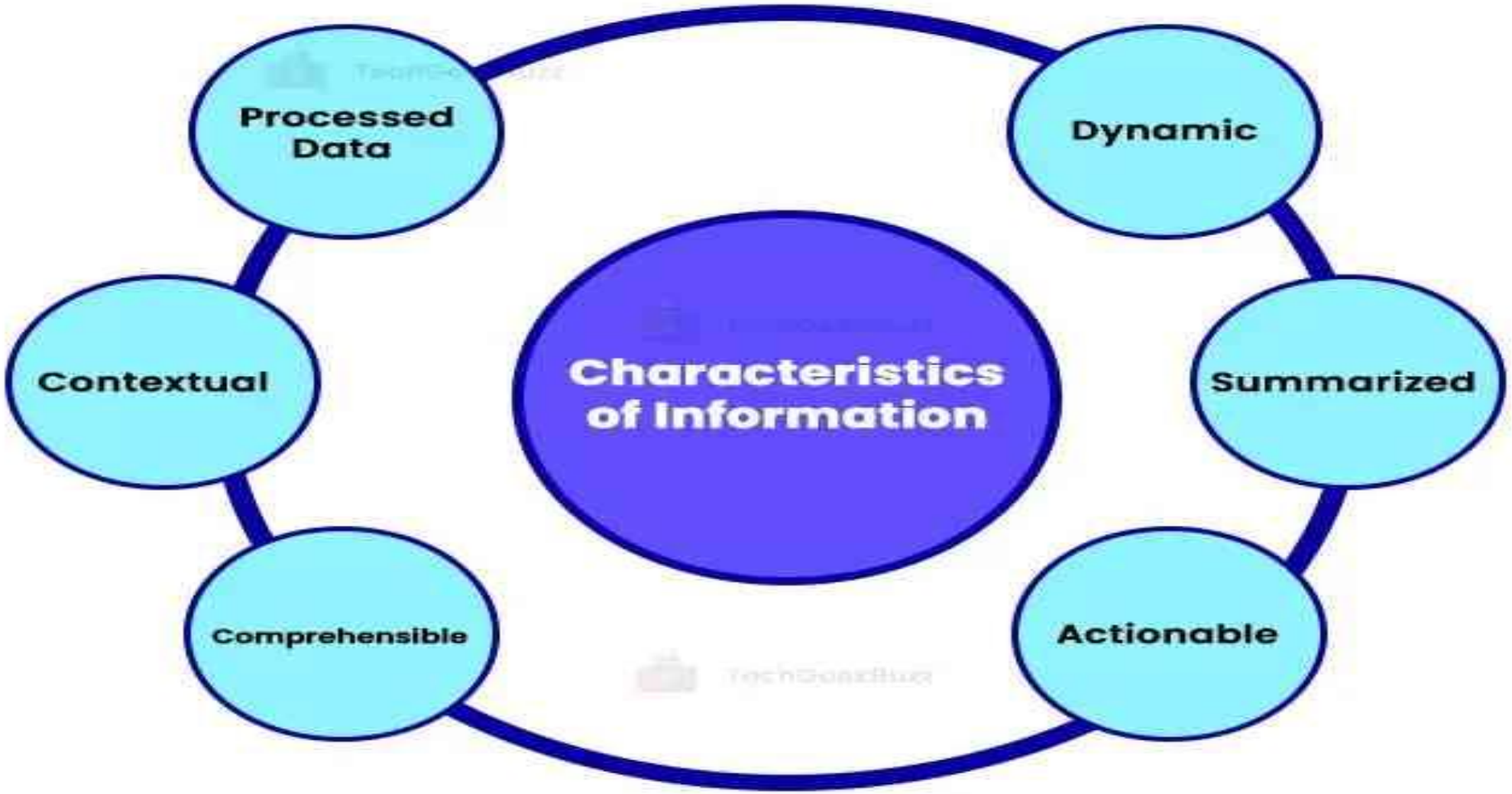
Data

- **Raw and Unprocessed:** Data is raw and disordered and does not have any meaning or context of its own. It simply collects figures, statistics, and facts in their original form.
- **No Objective:** Data does not have any purpose or significance of its own until it undergoes processing and interpretation.
- **Immutable:** It is impossible to modify data without adding new inputs or observations.
- **Quantitative or Qualitative:** We can represent data in numbers (quantitative) as well as in text, images, audio, and videos (qualitative).

Information

Information : Information is defined as structured, organized, and processed data, presented within a context that makes it relevant and useful to the person who needs it.

Information



Information

- **Processed Data:** Information is the processed raw data that provides relevance, context, and meaning. Contextual: It is related to a certain context and serves a specific purpose.
- **Comprehensible:** Humans can quickly grasp information (knowledge or insights) and use it in decision-making.
- **Actionable:** Information is actionable and practical, and it is useful in the problem-solving process.
- **Summarized:** Information highlights key insights from the processed data. Hence, it is summarized and concise.
- **Dynamic:** It keeps updating continuously as new data is processed.

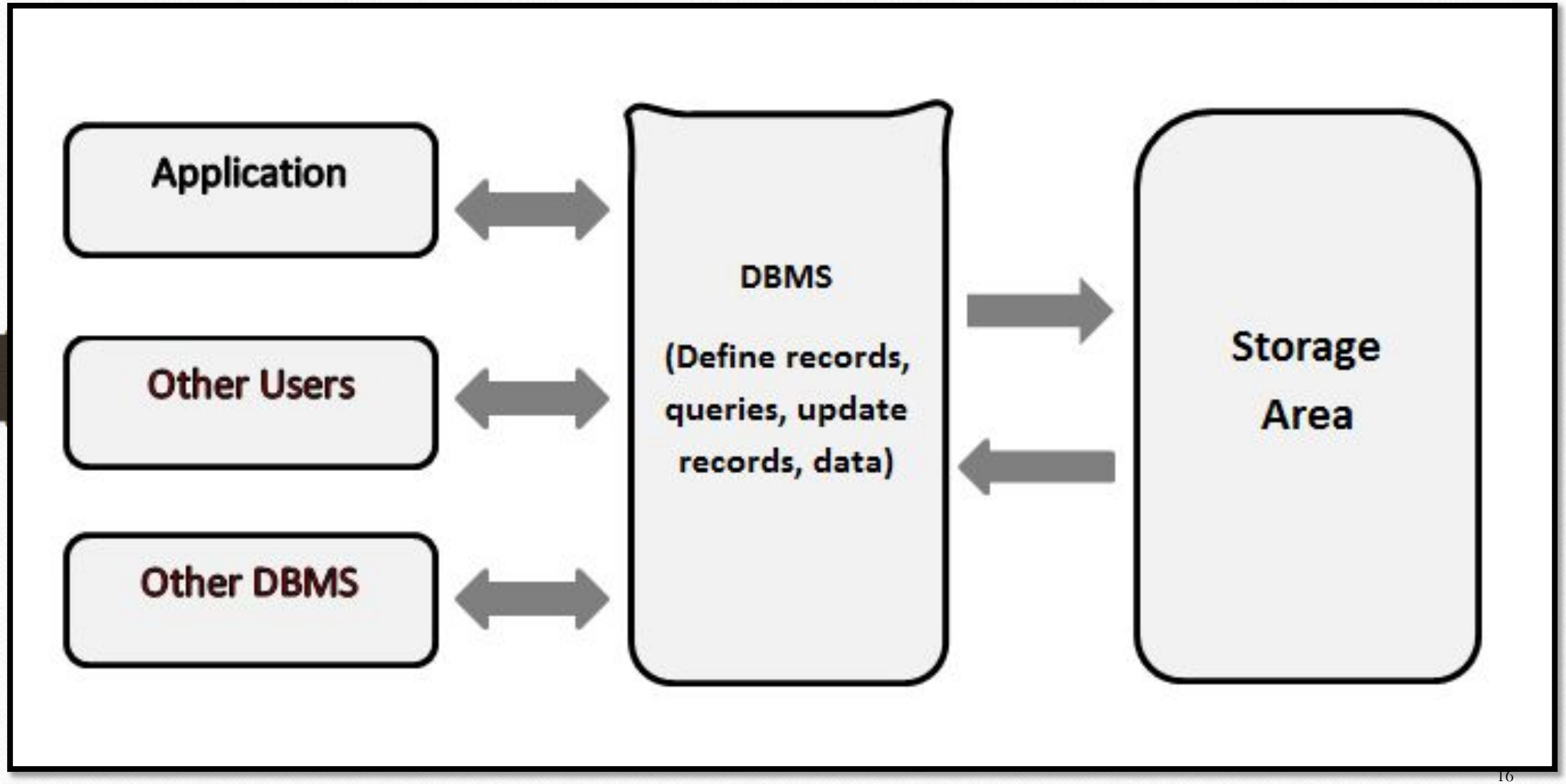
Introducing the Database

A database is an organized collection of structured information, or data, stored electronically in a computer system and managed by a database management system (DBMS).

What is a database?

- Organized collection
- Electronic storage
- Managed by DBMS
- Purpose-driven

Introducing the Database

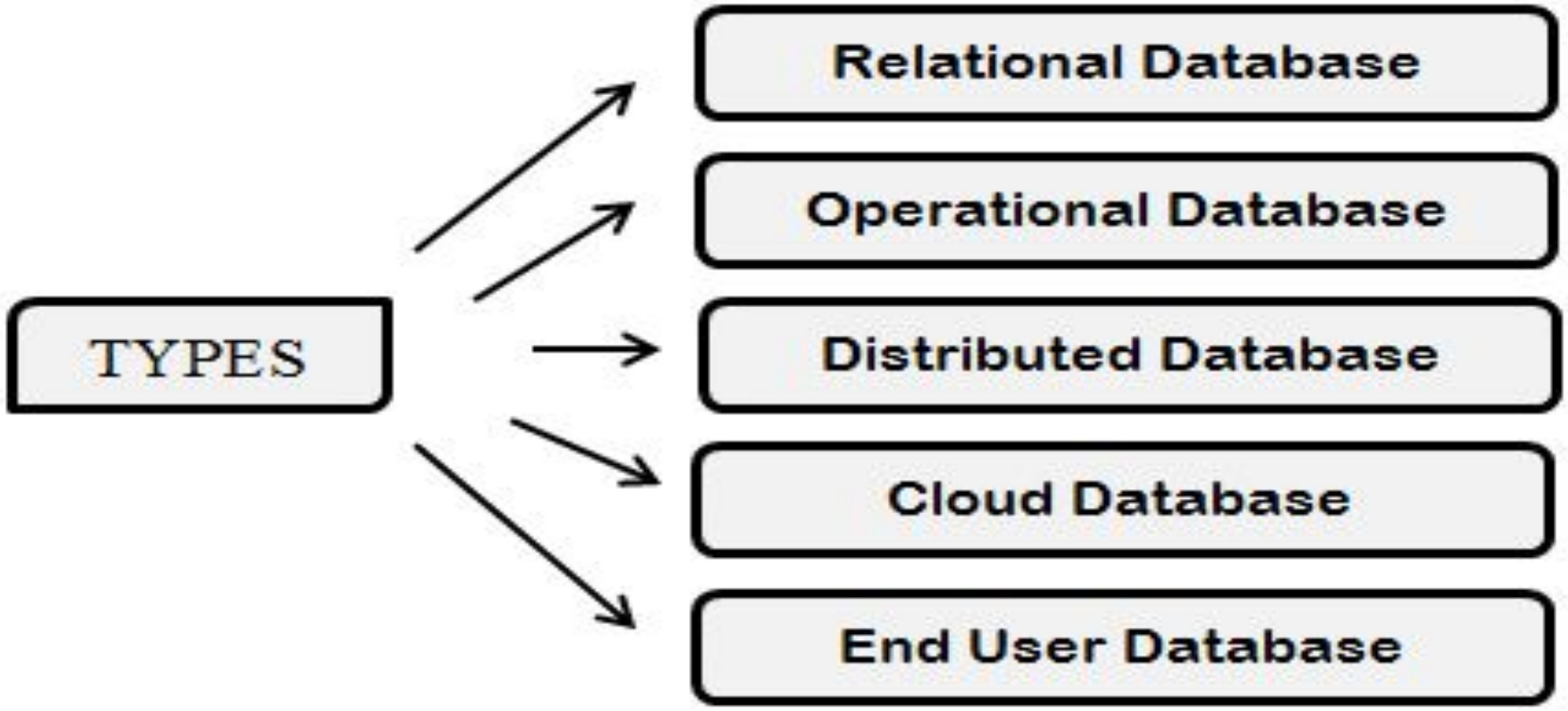


Types of Databases

- Centralized database
- Distributed database
- Personal database
- End-user database
- Commercial database
- NoSQL database

- Operational database
- Relational database
- Cloud database
- Object-oriented database
- Graph database

Types of Databases



Advantages of using Databases

- Reduced data redundancy
- Reduced updating errors and increased consistency
- Greater data integrity and independence from application programs
- Improved data access to users through the use of host and query languages
- Improved data security
- Reduced data entry, storage, and retrieval costs

File System

- A file system is an operating system component that manages how data is stored, organized, and retrieved on a storage medium (e.g., hard drive, SSD).
- It provides a hierarchical structure of files and directories, allowing users and applications to interact with data in a structured manner.

Key Features of a File System

Data Organization: Arranges data into files and directories (folders).

File Naming and Metadata: Assigns names to files and stores metadata like size, creation date, and permissions.

Storage Management: Handles the physical storage and retrieval of data blocks on the storage device.

Access Control: Implements permissions to regulate access to files and directories.

Basic File Terminology

Basic File Terminology

TERM	DEFINITION
Data	"Raw" facts, such as a telephone number, a birth date, a customer name, and a year-to-date (YTD) sales value. Data have little meaning unless they have been organized in some logical manner.
Field	A character or group of characters (alphabetic or numeric) that has a specific meaning. A field is used to define and store data.
Record	A logically connected set of one or more fields that describes a person, place, or thing. For example, the fields that constitute a record for a customer might consist of the customer's name, address, phone number, date of birth, credit limit, and unpaid balance.
File	A collection of related records. For example, a file might contain data about the students currently enrolled at Gigantic University.

Manual File System

- Historically, such systems were often manual, paper-and-pencil systems.
- The papers within these systems were organized in order to facilitate the expected use of the data.

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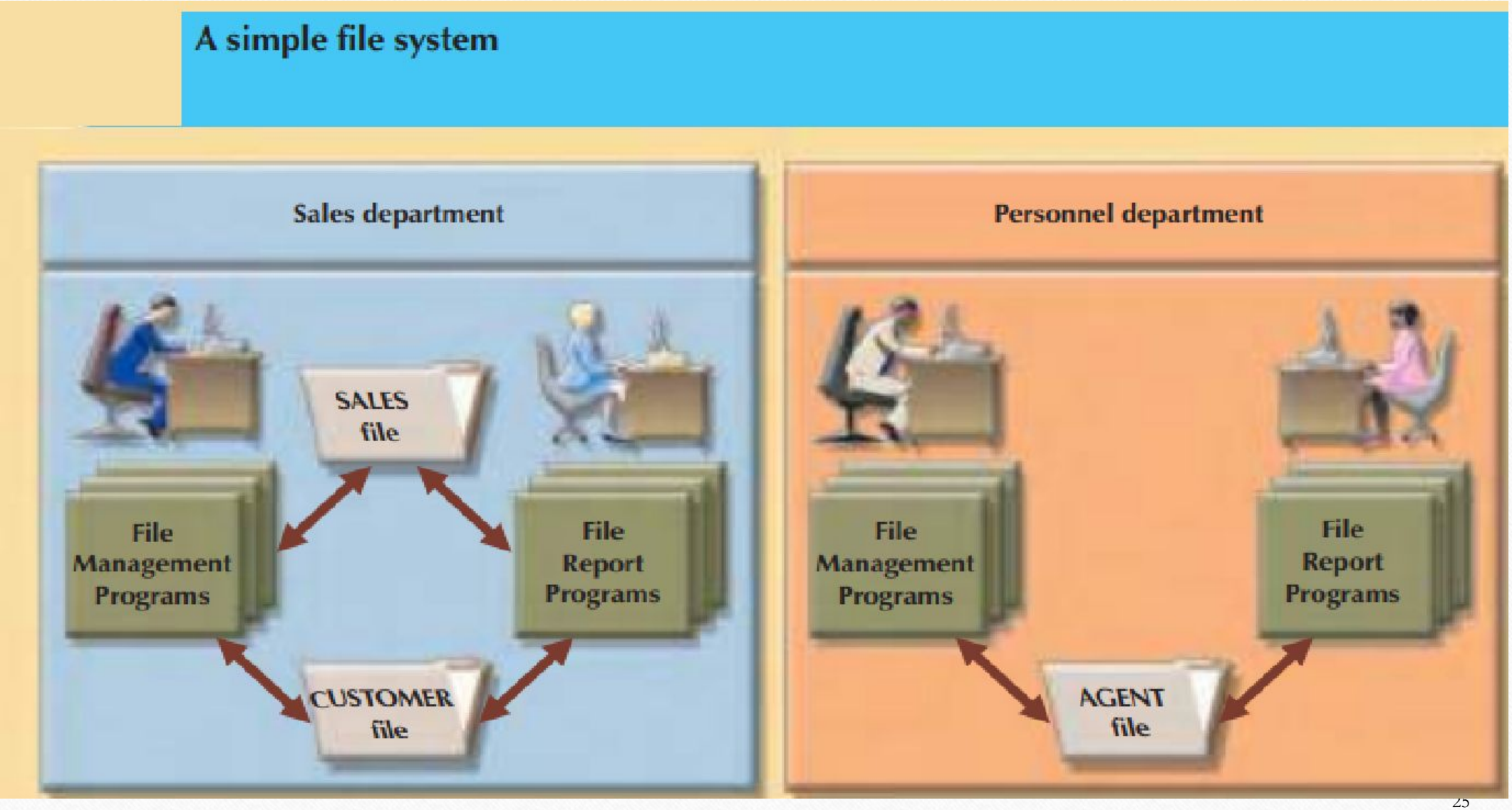
Computerized File System



Role of File System in DBMS

- Data Abstraction
- Data Integrity
- Data Redundancy Control
- Concurrency Control
- Security and Recovery
- Querying and Manipulation

Role of File System in DBMS

Basis	DBMS	File System
Definition	DBMS is a collection of data. Over here, the user is not required to write the procedures.	The file system is a collection of data, Where the user has to write the procedures which can lead to loss of data also.
Data Abstraction	It gives the abstract view of data that hides the details.	It provides the details of data representation and storage of data.
Crash Mechanism	DBMS provides the crash mechanism, i.e., DBMS protects the user from system failure.	In a file system, it doesn't have a crash recovery mechanism.
Protection	DBMS provides a good protection mechanism.	It is tough to protect a file system.

Role of File System in DBMS

Basis	DBMS	File System
Data Storing	It contains a wide variety of sophisticated technology to store and retrieve the data.	It can't efficiently store and retrieve the data.
Data Concurrency	It takes care of concurrent access of data using some form of locking.	Concurrent access in file system has many problems, like redirecting the file while deleting some information and updating some data.
Cost	It has a higher cost than the file system.	It is less expensive than DBMS.

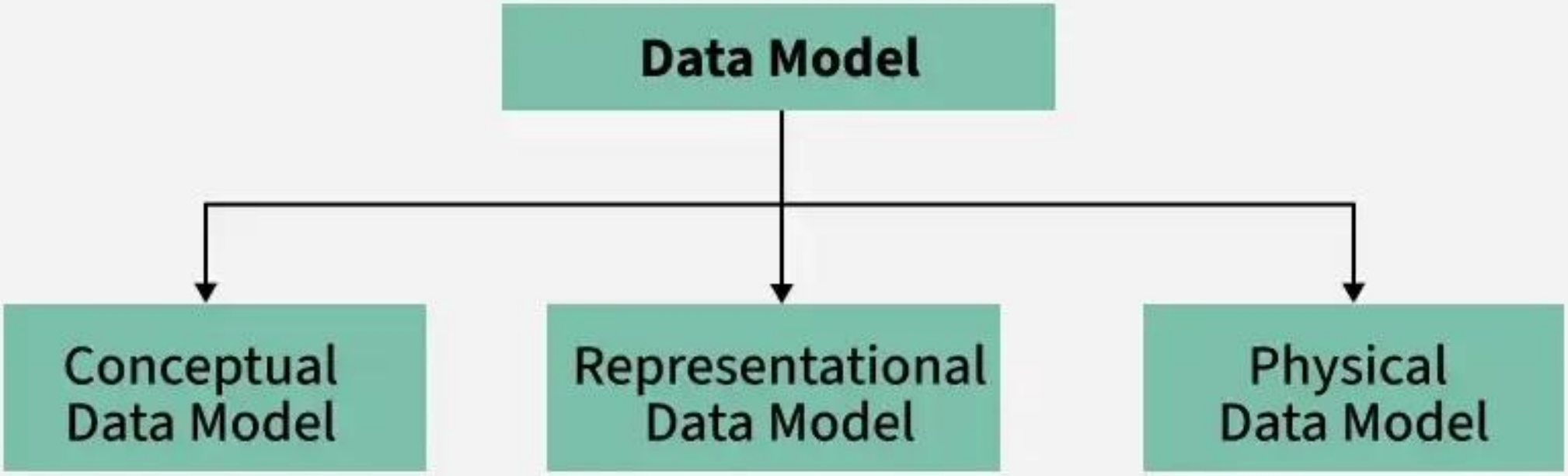
Problems of File System

- Lengthy development times
- Difficulty of getting quick answers
- Complex system administration
- Lack of security and limited data sharing
- Extensive programming. Making changes to an existing file structure can be difficult in a file system environment

DATA MODEL

- Data model is a collection of conceptual tools for describing data, data relationships, data semantics and consistency constraints.
- That means a data model provides a way to describe the design of a database.

CATEGORIES OF DATA MODELS



High Level or Conceptual Data Model

- The conceptual data model describes the database at a very high level and is useful to understand the needs or requirements of the database.
- It is this model, that is used in the requirement-gathering process i.e. before the Database Designers start making a particular database.
- One such popular model is the entity/relationship model (ER model).

Entity-Relationship Model (ER Model)

- It is a high-level data model which is used to define the data and the relationships between them.
- It is basically a conceptual design of any database which is easy to design the view of data.

Components of ER Model

- **Entity:** An entity is referred to as a real-world object. It can be a name, place, object, class, etc. These are represented by a rectangle in an ER Diagram.
- **Attributes:** An attribute can be defined as the description of the entity. These are represented by Ellipse in an ER Diagram. It can be Age, Roll Number, or Marks for a Student.
- **Relationship:** Relationships are used to define relations among different entities. Diamonds and Rhombus are used to show Relationships

Representational Data Model

- This type of data model is used to represent only the logical part of the database and does not represent the physical structure of the database.
- The representational data model allows us to focus primarily, on the design part of the database.
- A popular representational model is a Relational model. The relational Model consists of Relational Algebra and Relational Calculus.

Low Level or Physical Data Model

- It provide concepts that describe the details of how data is stored on the computer storage media, typically magnetic disks.
- Physical data models describe how data is stored as files in the computer by representing information such as record formats, record orderings, and access paths.

Importance of Data Models

- **Organization**: Structures data logically, making it easier to understand and manage.
- **Data Quality**: Helps identify and rectify inconsistencies, leading to better data quality and accuracy.
- **Efficiency**: Eliminates redundancy and enables efficient data retrieval.
- **Communication**: Serves as a communication tool between designers, programmers, and users.
- **Single Source of Truth**: Establishes a unified, consistent view of data across the organization.

Basic Building Blocks

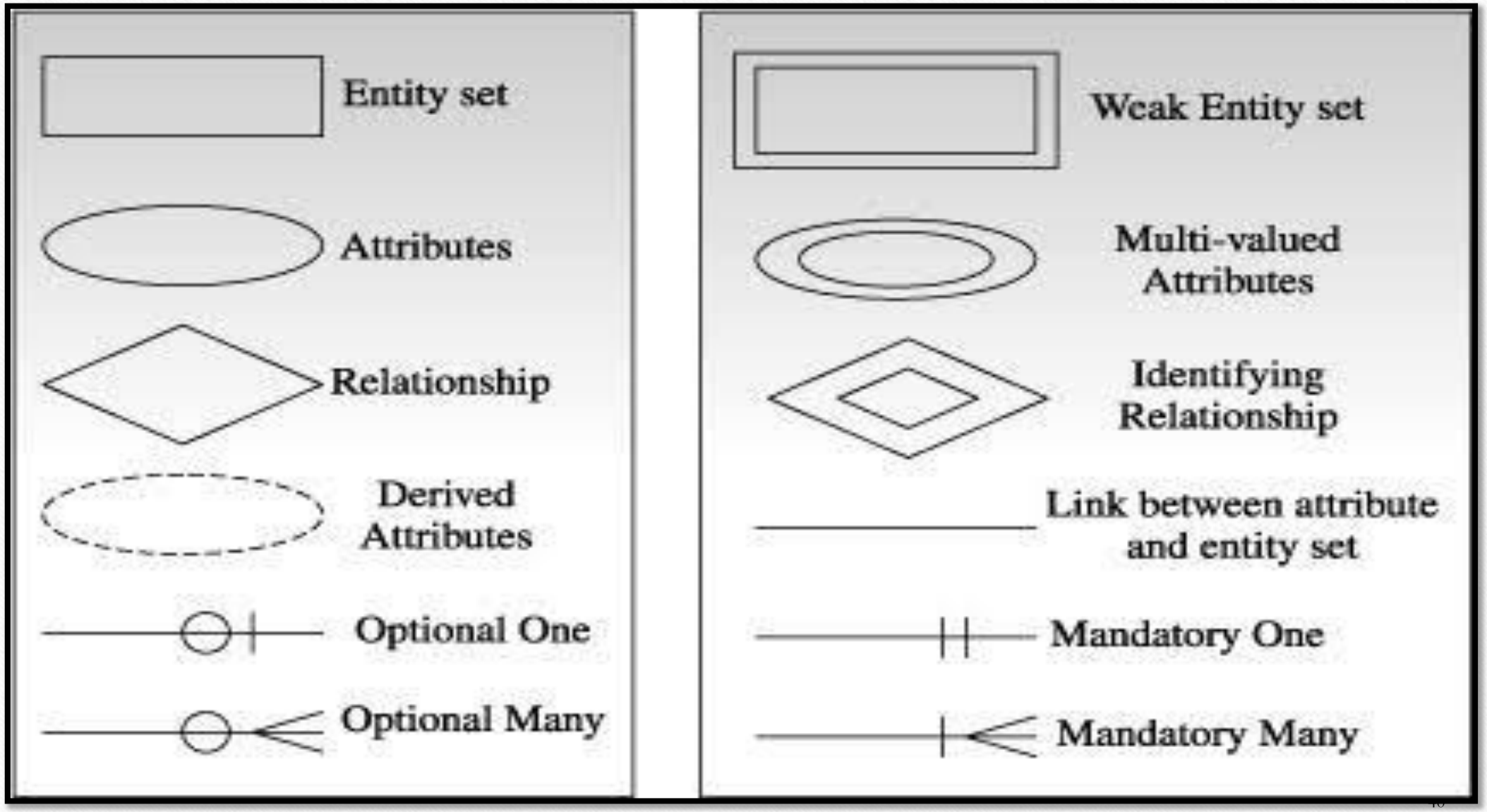
- **Entities**: Real-world objects or concepts (e.g., "Customer," "Product").
- **Attributes**: Characteristics or properties of an entity (e.g., for "Customer," attributes could be name, address).
- **Relationships**: Associations between entities (e.g., a "Customer" places an "Order").
- **Constraints**: Rules that ensure data integrity (e.g., a value must be unique or within a certain range).
- **Keys**: Special attributes used to identify records uniquely.

Basic Building Blocks

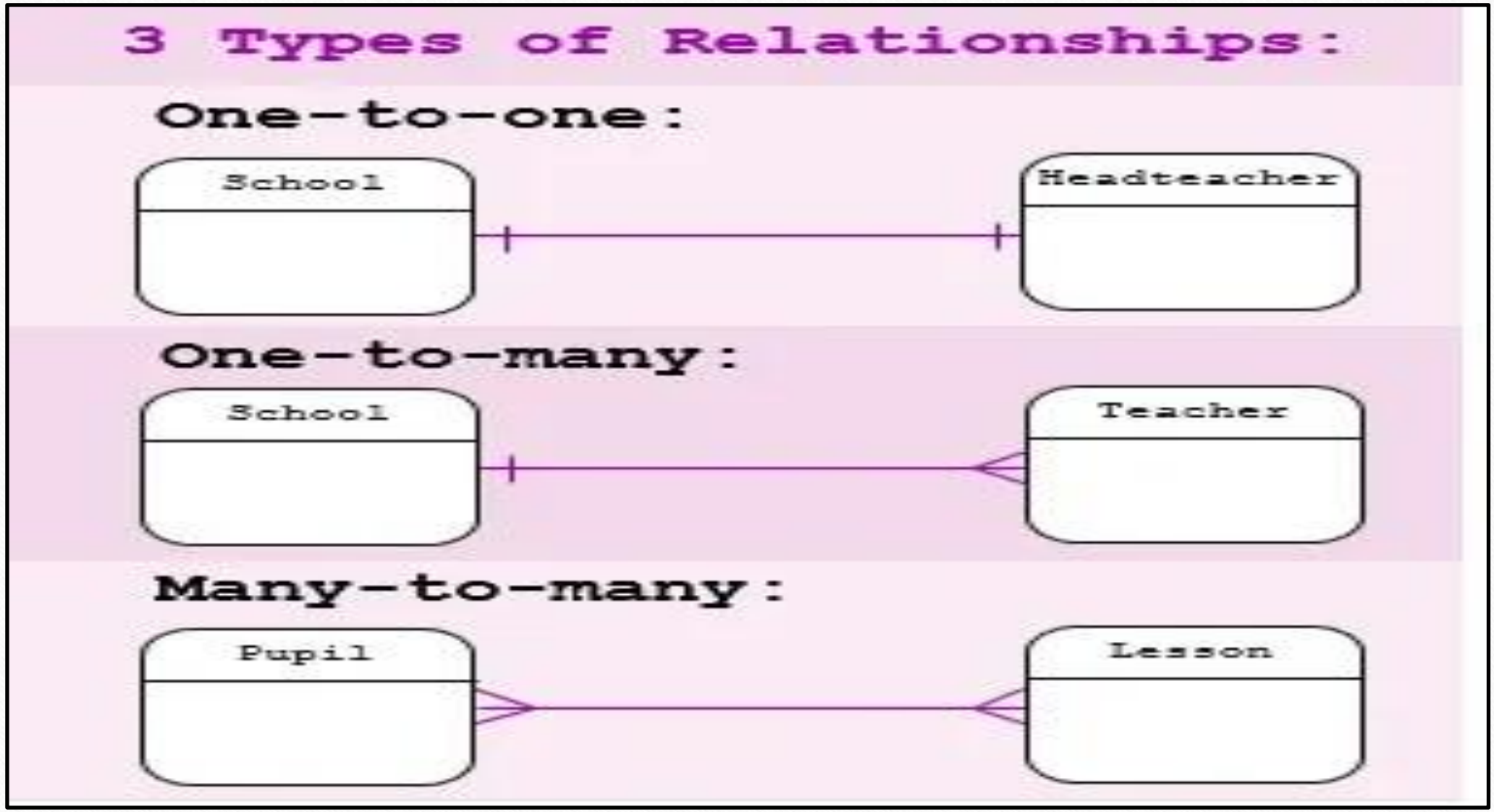
- **Cardinality:** The term "cardinality" refers to the maximum number of times an entity can be related to a single instance of another entity in a relation.
- **Primary Key:** A primary key is a unique identifier for an entity in a database. It is used to differentiate between different instances of the same entity.

In the ER model, a primary key is represented by an underline beneath an attribute oval. For example, the "student ID" attribute of the "student" entity would be a primary key.

Basic Building Blocks



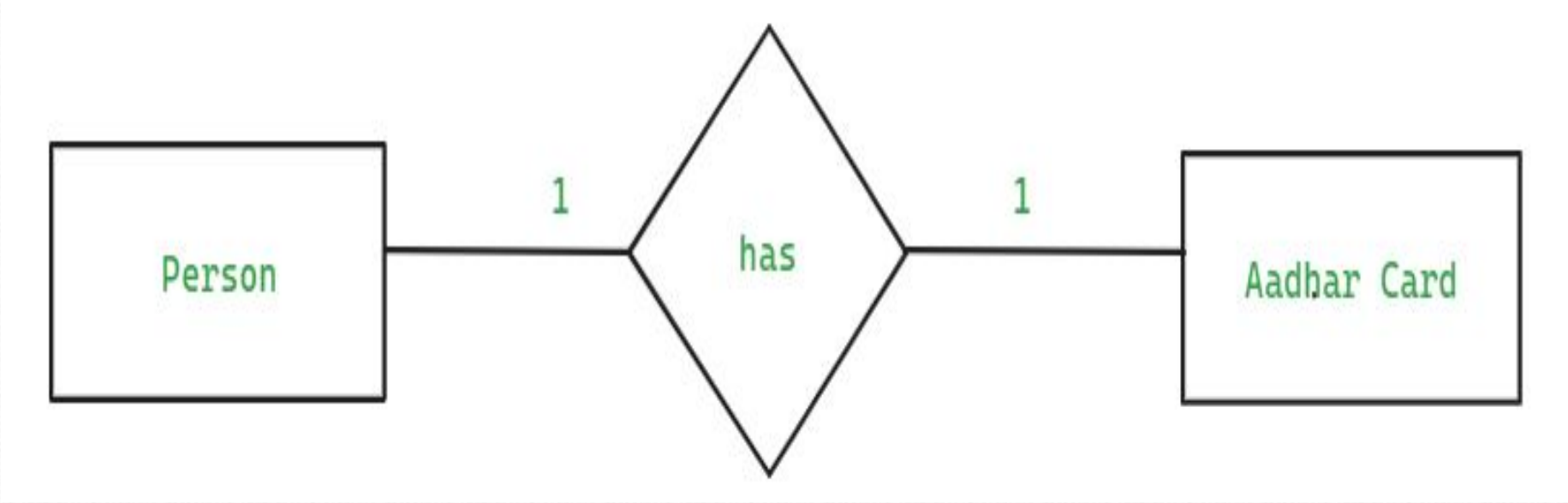
Types of Relationships



One to One Relationship

In a one-to-one relationship, each record in one table is linked to a single record in another table.

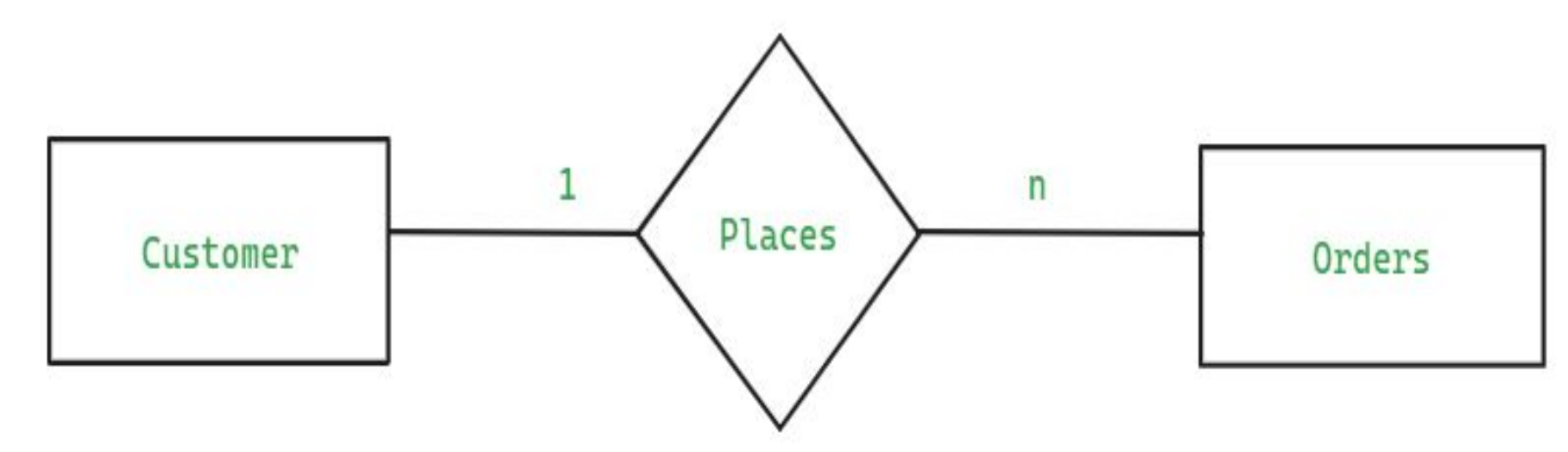
One-to-Many (1:1) Relationship



One to Many or or Many-to-One Relationship

- In one-to-many or many-to-one relationship, a single record in one table can be associated with multiple records in another table and this is the most common type of relationship in DBMS.

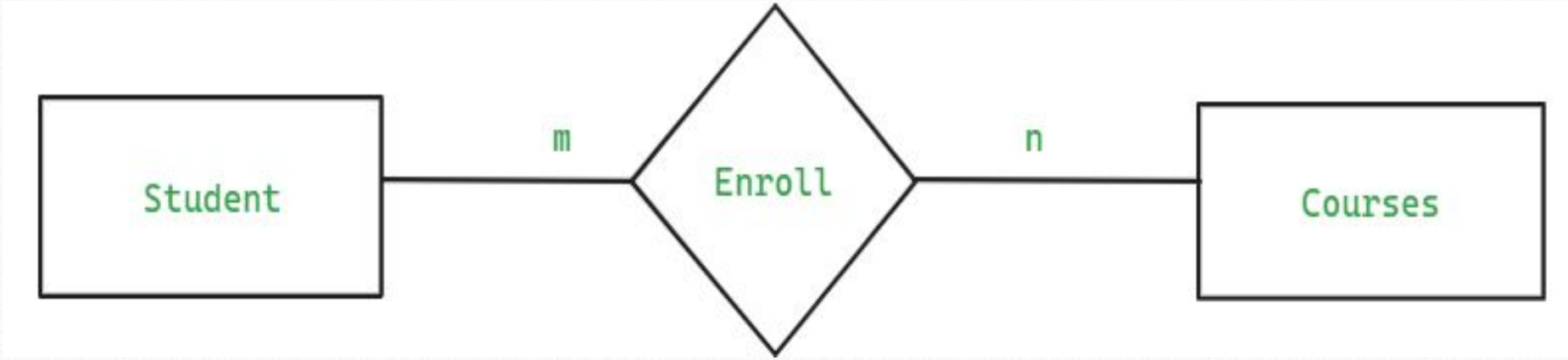
One-to-Many (1:M) Relationship



Many-to- Many Relationship

- A many-to-many relationship is relationship in which one multiple records in one table are associated with multiple records in another table. This relationship is mainly implemented using junction table.

Many-to-Many (M:N) Relationship



Business Rules

- Business rules, derived from a detailed description of an organization's operations, help to create and enforce actions within that organization's environment.
- Business rules must be rendered in writing and updated to reflect any change in the organization's operational environment.

Discovering Business Rules

- They help to standardize the company's view of data.
- They can be a communications tool between users and designers.
- They allow the designer to understand the nature, role, and scope of the data. They allow the designer to understand business processes.
- They allow the designer to develop appropriate relationship participation rules and constraints and to create an accurate data model.

Purpose of Business Rules

- **Ensure data integrity**: Rules like "every customer must have a name" can be enforced with constraints like NOT NULL, ensuring data accuracy.
- **Reflect business policies**: They translate real-world organizational logic, such as "an onboarding date must not be earlier than the contract start date," into database constraints.

Purpose of Business Rules

- **Prevent user errors**: By defining valid data and relationships, rules prevent invalid entries and maintain consistency across the database.
- **Improve communication**: They serve as a tool for communication between business users and developers, ensuring the technical model aligns with business need

Evolution of Data Models

Evolution of Major Data Models

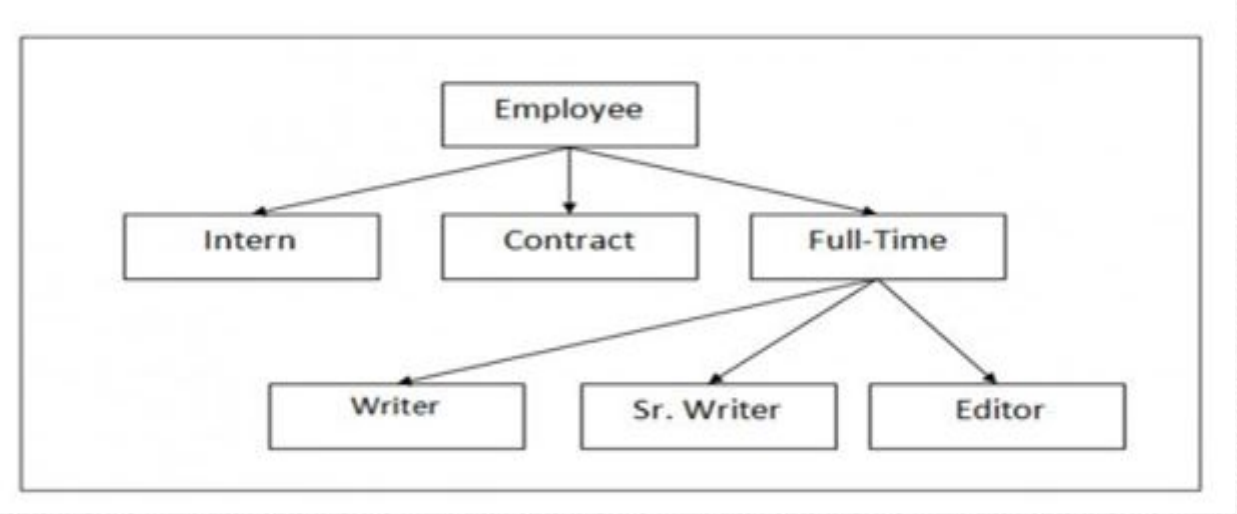
GENERATION	TIME	DATA MODEL	EXAMPLES	COMMENTS
First	1960s–1970s	File system	VMS/VSAM	Used mainly on IBM mainframe systems Managed records, not relationships
Second	1970s	Hierarchical and network	IMS ADABAS IDS-II	Early database systems Navigational access
Third	Mid-1970s to present	Relational	DB2 Oracle MS SQL-Server MySQL	Conceptual simplicity Entity relationship (ER) modeling and support for relational data modeling
Fourth	Mid-1980s to present	Object-oriented Object/relational (O/R)	Versant Objectivity/DB DB/2 UDB Oracle 11g	Object/relational supports object data types Star Schema support for data warehousing Web databases become common
Next generation	Present to future	XML Hybrid DBMS	dbXML Tamino DB2 UDB Oracle 11g MS SQL Server	Unstructured data support O/R model supports XML documents Hybrid DBMS adds an object front end to relational databases

Evolution of Data Models

- **Hierarchical and Network Models:** Early models from the 1960s that represented data in a tree like or graph-like structure.
- **Relational Model:** Introduced in the late 1960s, this model organizes data into tables with rows and columns, becoming the most widely used model for decades.
- **Entity-Relationship Model:** A conceptual model that uses entities and relationships to represent data, often used as a blueprint for relational databases.
- **Object-Oriented Models**: Models that combine database and object-oriented programming concepts.

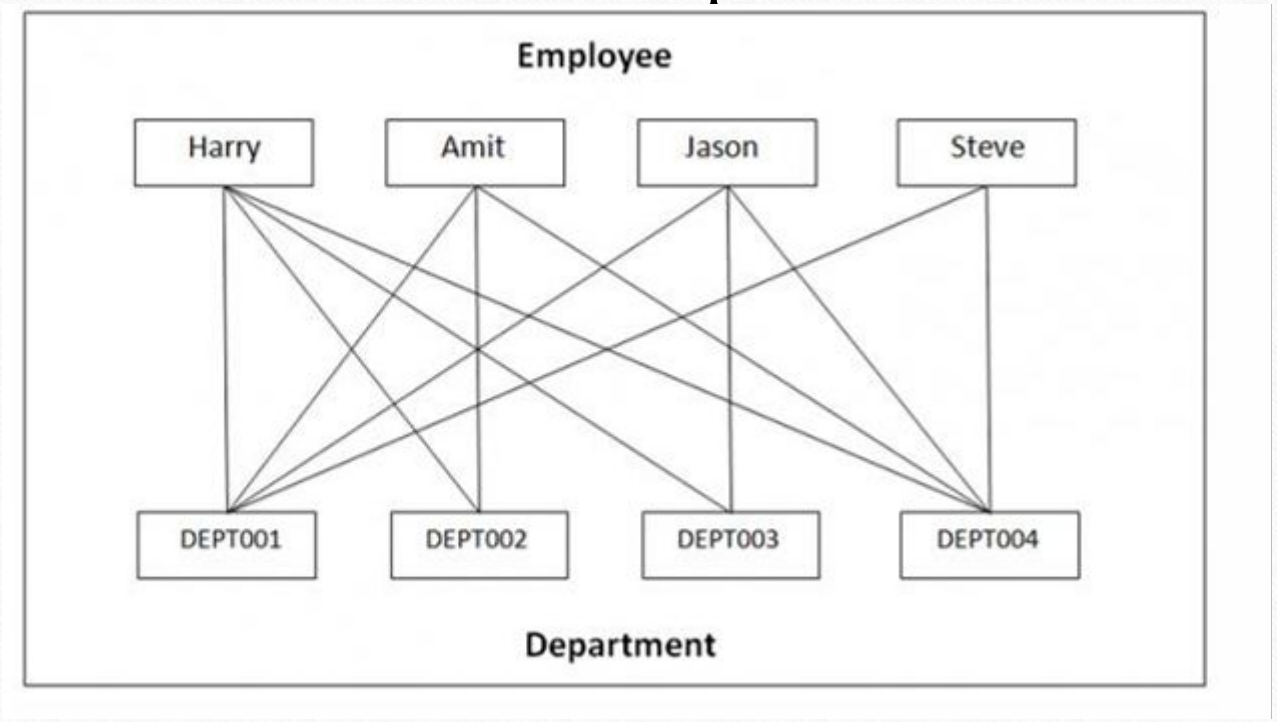
Hierarchical Model

- The hierarchical model classifies the data into a tree-like structure with a single parent or root for each record.
- Sibling records are arranged in a certain sequence.
- This is the physical order in which the database is stored.
- This model is useful for explaining a wide range of real world interactions.



Network Model

- In the network model, the user perceives the network database as a collection of records in 1:M relationships.
- However, unlike the hierarchical model, the network model allows a record to have more than one parent.



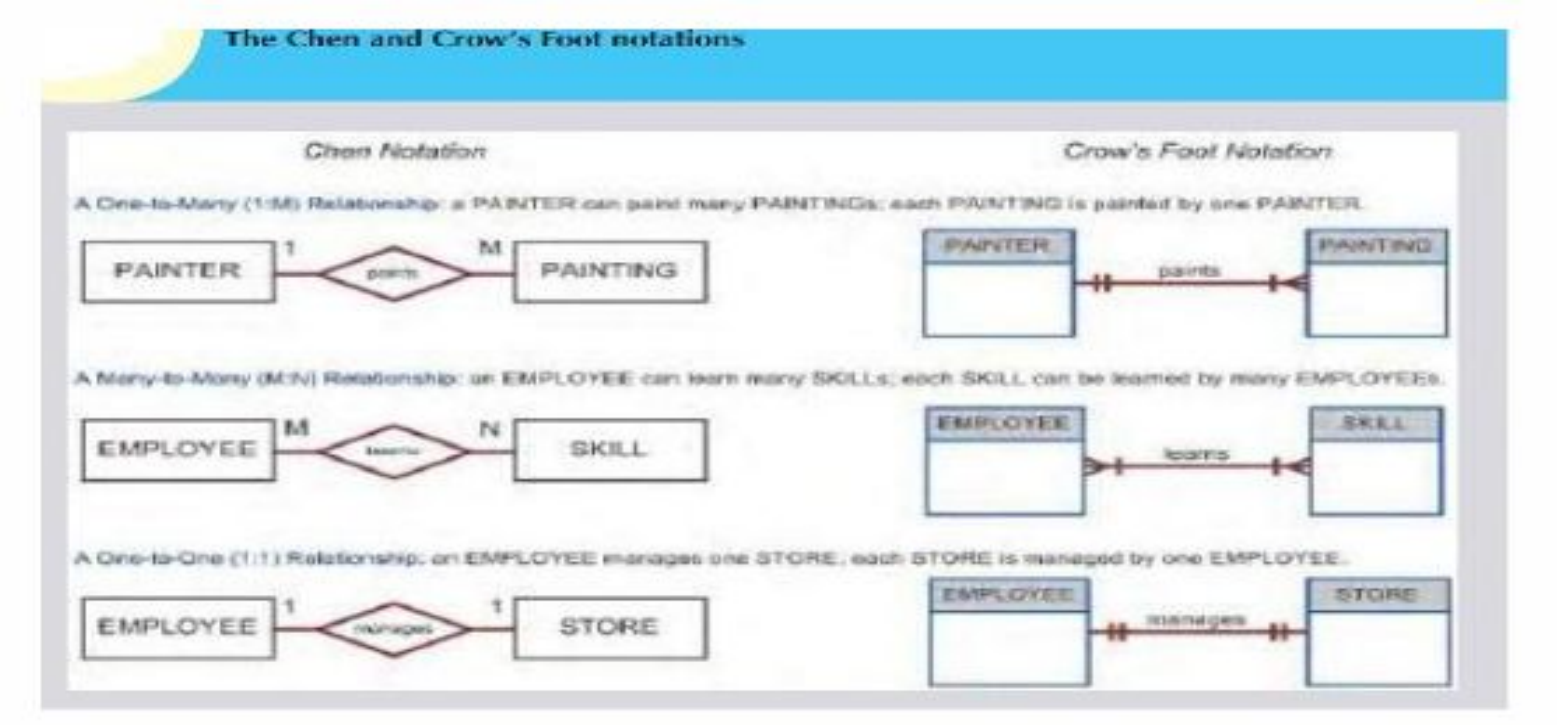
Relational Model

- A relational model groups data into one or more tables. These tables are related to each other using common records.
- The data is represented in the form of rows and columns i.e. tables:

	Column1	Column2	Column3
Row1			
Row2			
Row3			
Row4			
Row5			

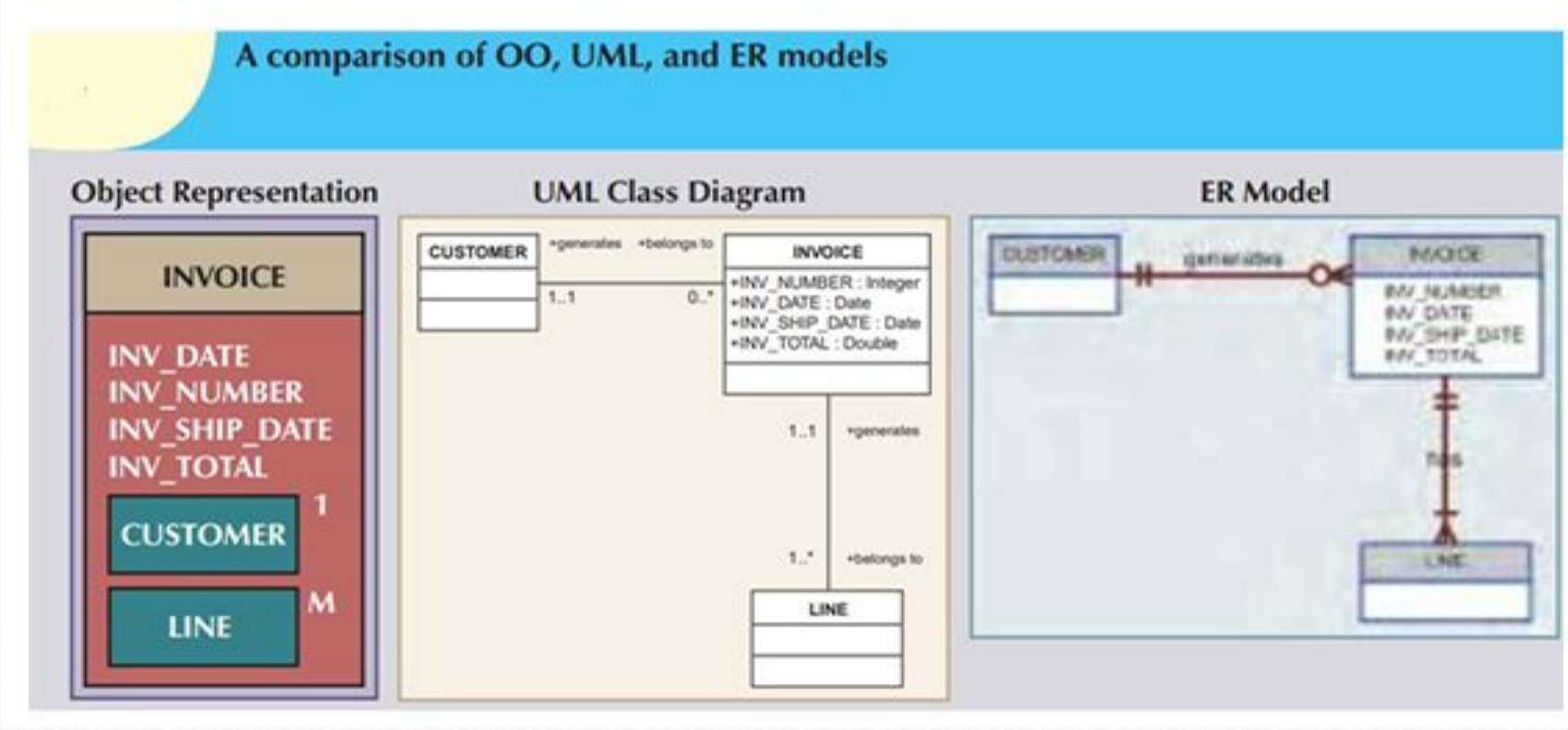
Entity Relationship (ER) Model

- Different types of relationships using two ER notations: the original Chen notation and the more current Crow's Foot notation

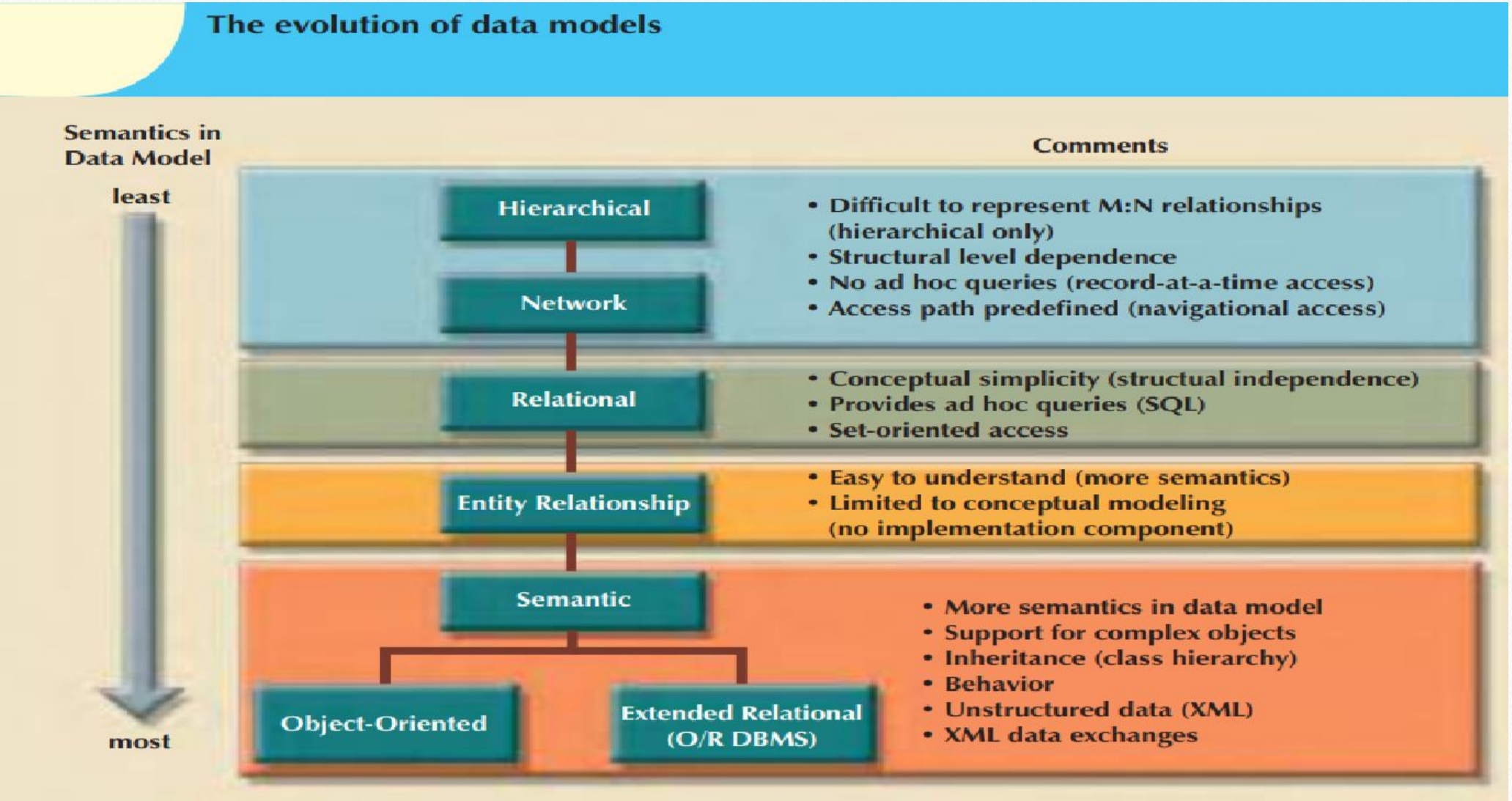


The Object- Oriented (OO) Model

- Different types of relationships using two ER notations: the original Chen notation and the more current Crow's Foot notation



Evolution of Data Models



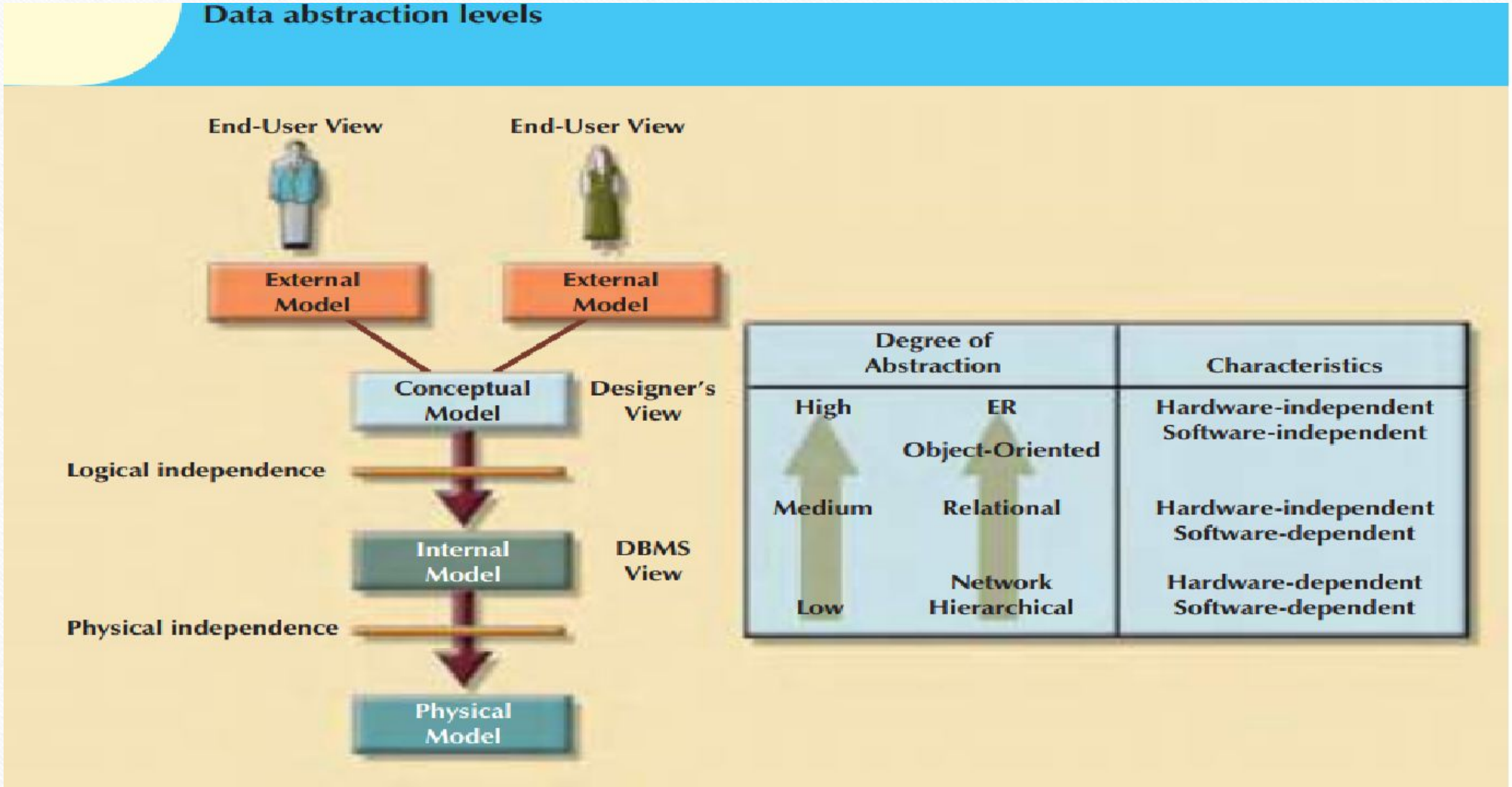
Data Abstraction

- Data abstraction is the process of hiding unwanted and irrelevant details from the end user.
- It helps to store information in such a way that the end user can access data which is necessary, the user will not be able to see what data is stored or how it is stored in a database.

Data Abstraction

- Data abstraction hides unwanted details, showing users only relevant data.
- It keeps data safe by hiding how and where data is stored.
- Abstraction simplifies access while maintaining data security and efficiency.

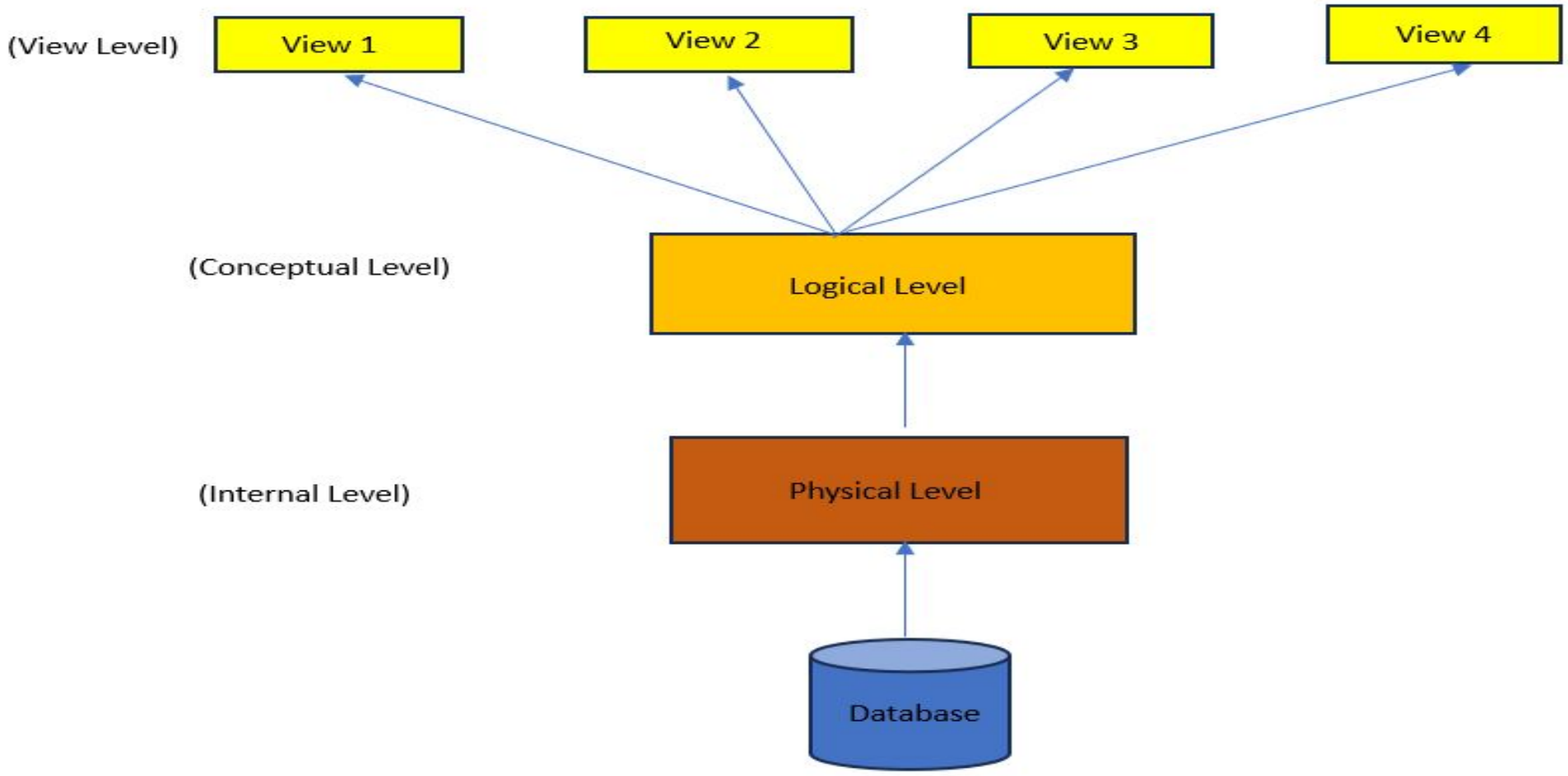
Data Abstraction Levels



Degrees of Data Abstraction

- There are three degrees of data abstraction in DBMS: **Physical (Internal)**, **Logical (Conceptual)**, and **View (External)**.
- These levels simplify database interaction by hiding unnecessary details, from the lowest level of how data is physically stored to the highest level of how users view the data.

Degrees of Data Abstraction



Degrees of Data Abstraction

- **Physical (Internal) Level:** This is the lowest level, describing how data is actually stored on the hardware, including details like file structures, indexes, and data block addresses.
- **Logical (Conceptual) Level:** This is the middle level, which describes what data is stored in the database and the relationships between different data items, without mentioning the physical storage details.

Degrees of Data Abstraction

- **View (External) Level:** This is the highest level, providing different views of the data to different users. Each view shows only the relevant information for a specific user, making the database easier to use and more secure.

Question Bank

Section A

1. Which of the following best defines a database?

A. A collection of unrelated files
B. A collection of interrelated data stored with minimum redundancy
C. A collection of unstructured information
D. A type of programming language
Answer: B

2 **In traditional file systems, which problem is common?**

A. High data abstraction
B. Data independence
C. Data redundancy and inconsistency
D. Automatic integrity enforcement
Answer: C

Question Bank

3. A database management system (DBMS) primarily helps in:

A. Designing operating systems
B. Managing data efficiently
C. Running hardware devices
D. Writing code for applications
Answer: B

4. In traditional file systems, which problem is common?

A. High data abstraction
B. Data independence
C. Data redundancy and inconsistency
D. Automatic integrity enforcement
Answer: C

Question Bank

5. Business rules in data modeling help to:

A. Increase redundancy
B. Define relationships and constraints
C. Replace the need for a database
D. Convert files into tables automatically
Answer: B

6. In traditional file systems, which problem is common?

A. High data abstraction
B. Data independence
C. Data redundancy and inconsistency
D. Automatic integrity enforcement
Answer: C

Question Bank

7. Which of the following is the simplest level of data abstraction?

A. Logical level
B. View level
C. Physical level
D. Conceptual level
Answer: C

8. In traditional file systems, which problem is common?

A. High data abstraction
B. Data independence
C. Data redundancy and inconsistency
D. Automatic integrity enforcement
Answer: C

Question Bank

9. Which is NOT a basic building block of a data model?

A. Entities
B. Attributes
C. Relationships
D. Algorithms
Answer: D

10. The relational model was introduced by:

A. Bill Gates
B. Edgar F. Codd
C. Charles Babbage
D. Dennis Ritchie
Answer: B

Question Bank

Section B

1. Describe the components of a DBMS with suitable examples.
2. Differentiate between data and information with suitable examples.
3. Define the term “database.” What are its essential characteristics?
4. Describe the traditional file system approach. How does it store and manage data?
5. Explain the basic building blocks of data models: entities, attributes, relationships, and constraints.

Question Bank

Section C

1. Explain the main characteristics of a database system
2. Explain the major problems of the file system approach. (Data redundancy, inconsistency, poor security, integrity issues)
3. What is a data model? Describe its purpose in database design
4. Compare different data models: hierarchical, network, relational, and object-oriented models.
5. Describe the three levels of data abstraction: physical, logical, and external.

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Online Tools

1.sqlfiddle.com

2.Draw.io